ANNIHILATORS OF SECTIONS GO DOWN ALONG FLAT LOCALLY QUASI-FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12932-12976, immediately following the statement at 12913-12930.

Editorial note. The proof constructs psi and the map A[I]^{r} -> B[I] explicitly; both are displayed inside the span.

Editorial note. Three cross-chapter/same-chapter pointers (Algebra, Modules) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Stratification by fibre degree in the quasi-finite case. chapter lemma lemma-stratify-flat-fp-qf, cited in the proof. Provides the stratification by fibre degree used to reduce to a finite locally free piece.
- Uniformly bounded stratification. chapter lemma lemma-stratify-flat-fp-lqf-universally-bounded, cited in the proof. Supplies the boundedness of the degree over the stratum, which is what makes the rank r constant there.
- Support of a module and quasi-compact opens. Modules, definition modules-definition-support and Algebra, lemma algebra-lemma-qc-open, cited in the proof. Controls the support of the section and lets the construction be restricted to quasi-compact opens.
- The ideal A[I] of elements annihilating I. construction used at proof lines 12966-12975. Produces the quasi-coherent subsheaf F and the map F^{r} -> pi_*O_V realising the annihilator on the base.

Source reference. Original reference 00F6: focused-seeds/geometry/S9-morph-go-down-annihilators/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07RY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12761–12761. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07RZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12845–12845. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01AT: focused-seeds/geometry/S9-morph-go-down-annihilators/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is flat, locally of finite presentation, and locally quasi-finite. Let $g \in \Gamma(X, \mathcal{O}_X)$ nonzero. Then there exist an open $V \subset X$ such that $g|_V \neq 0$, an open $U \subset S$ fitting into a commutative diagram*

$$\begin{array}{ccc} V & \longrightarrow & X \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle f} \\ U & \longrightarrow & S, \end{array}$$

a quasi-coherent subsheaf $\mathcal{F} \subset \mathcal{O}_U$, an integer $r > 0$, and an injective $\mathcal{O}_U$-module map $\mathcal{F}^{\oplus r} \to \pi_*\mathcal{O}_V$ whose image contains $g|_V$.

Proof. We may assume X and S affine. We obtain a filtration $\emptyset = Z_{-1} \subset Z_0 \subset Z_1 \subset Z_2 \subset \ldots \subset Z_n = S$ as in Lemmas 07RY and 07RZ. Let $T \subset X$ be the scheme theoretic support of the finite $\mathcal{O}_X$-module $\mathrm{Im}(g : \mathcal{O}_X \to \mathcal{O}_X)$. Note that T is the support of g as a section of $\mathcal{O}_X$ (Modules, Definition 01AT) and for any open $V \subset X$ we have $g|_V \neq 0$ if and only if $V \cap T \neq \emptyset$. Let r be the smallest integer such that $f(T) \subset Z_r$ set theoretically. Let $\xi \in T$ be a generic point of an irreducible component of T such that $f(\xi) \notin Z_{r-1}$ (and hence $f(\xi) \in Z_r$). We may replace S by an affine neighbourhood of $f(\xi)$ contained in $S \setminus Z_{r-1}$. Write $S = \mathrm{Spec}(A)$ and let $I = (a_1, \ldots, a_m) \subset A$ be a finitely generated ideal such that $V(I) = Z_r$ (set theoretically, see Algebra, Lemma 00F6). Since the support of g is contained in $f^{-1}V(I)$ by our choice of r we see that there exists an integer N such that $a_j^N g = 0$ for $j = 1, \ldots, m$. Replacing a_j by a_j^r we may assume that $Ig = 0$. For any A-module M write $M[I]$ for the I-torsion of M, i.e., $M[I] = \{m \in M \mid Im = 0\}$. Write $X = \mathrm{Spec}(B)$, so $g \in B[I]$. Since $A \to B$ is flat we see that

$$B[I] = A[I] \otimes_A B \cong A[I] \otimes_{A/I} B/IB$$

By our choice of Z_r, the A/I-module B/IB is finite locally free of rank r. Hence after replacing S by a smaller affine open neighbourhood of $f(\xi)$ we may assume that $B/IB \cong (A/IA)^{\oplus r}$ as A/I-modules. Choose a map $\psi : A^{\oplus r} \to B$ which reduces modulo I to the isomorphism of the previous sentence. Then we see that the induced map

$$A[I]^{\oplus r} \longrightarrow B[I]$$

is an isomorphism. The lemma follows by taking $\mathcal{F}$ the quasi-coherent sheaf associated to the A-module $A[I]$ and the map $\mathcal{F}^{\oplus r} \to \pi_*\mathcal{O}_V$ the one corresponding to $A[I]^{\oplus r} \subset A^{\oplus r} \to B$. □